

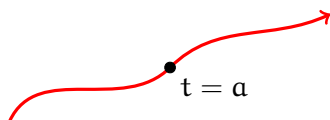
Example 1. Let $\mathcal{C}$ be the curve parametrized by:

$$\vec{r}(t) = \langle \sqrt{t}, 2 - t, t^2 \rangle$$

Find a parametrization for the **tangent line** to the curve at the point $P(2, -2, 16)$.

Solution. First we identify the value of the parameter that produces P .

Here is a sketch of a tangent line to a parametrized curve at parameter $t = a$.



A direction vector for the tangent line at parameter $t = a$ is the:

In our case we calculate:

Thus a parametrization for the tangent line to the curve at P is

$$\vec{\ell}(s) =$$

where we use a different parameter s to avoid confusion with parameter t .

Example 2. Last class, you showed that the curves $\mathcal{C}_1$ and $\mathcal{C}_2$ parametrized by:

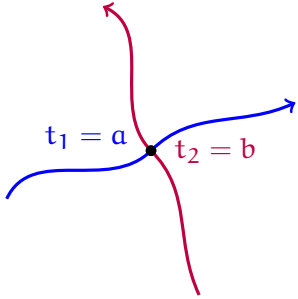
$$\vec{r}_1(t_1) = \langle t_1, 1 - t_1, 3 + t_1^2 \rangle$$

$$\vec{r}_2(t_2) = \langle 3 - t_2, t_2 - 2, t_2^2 \rangle$$

intersect at $\vec{r}_1(1) = \vec{r}_2(2) = \langle 1, 0, 4 \rangle$.

Find the **angle** between the paths at this point of intersection.

Solution. Here is sketch of parametrized curves and the angle between them.



The angle between the curves at their intersection is the angle between:

Since the curves intersect at $t_1 = 1$ and $t_2 = 2$, this is the angle between:

We present the two equivalent expressions of dot product, then solve for θ .

A. Path Derivative Rules. Path derivatives come equipped with rules.

Let $\vec{r}_1(t)$ and $\vec{r}_2(t)$ be differentiable paths.

Sum: $\left[\vec{r}_1(t) + \vec{r}_2(t) \right]' =$

Const. Mult: if c is constant, then $\left[c \vec{r}(t) \right]' =$

Dot Prod: $\left[\vec{r}_1(t) \cdot \vec{r}_2(t) \right]' =$

Cross Prod: $\left[\vec{r}_1(t) \times \vec{r}_2(t) \right]' =$

Example 3. Let $\vec{r}(t)$ differentially parametrize a curve lying on the unit sphere:

$$x^2 + y^2 + z^2 = 1$$

Show that its velocity vector $\vec{r}'(t)$ is always orthogonal to its position vector $\vec{r}(t)$.

Solution. Because $\vec{r}(t)$ is always a position vector for a point on the unit sphere

We rewrite this using dot products.

Next we take a derivative of both sides and apply the dot product rule.

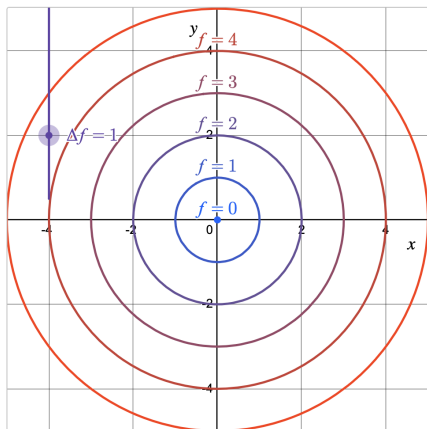
B. Multivariable Functions and Contours. Let's consider the function:

$$f(x, y) = \sqrt{x^2 + y^2}$$

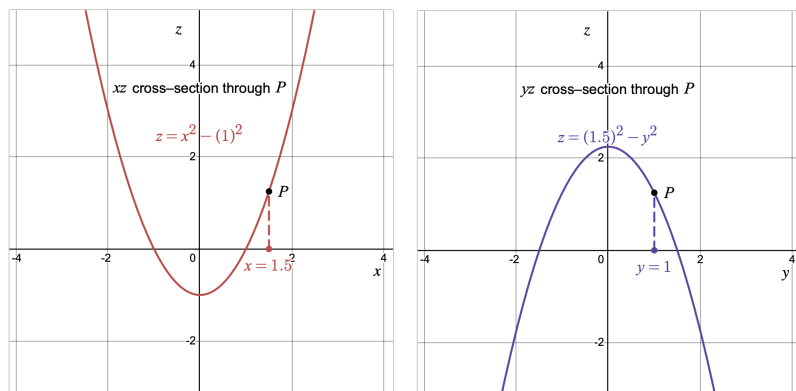
To graph a function, we set the outputs equal to another variable. In our case:

When this particular graph is sketched, the result is:

We sketch its **level curves** or **contours** by considering horizontal cross-sections.



C. Partial Derivatives. We consider slopes of graph $z = f(x, y)$ at given points.



For example let $f(x, y) = x^2 - y^2$ and **P** be the point on its graph with coordinates:

$$x = a, y = b, z =$$

The slope of the **xz**-cross section at **P** is the **x-partial derivative**:

We find it by holding **y** constant at value **b** and then taking an **x**-derivative:

$$f_x(x, b) =$$

$$\text{so: } f_x(a, b) =$$

The slope of the **yz**-cross section at **P** is the **y-partial derivative**:

We find it by holding **x** constant at value **a** and then taking an **y**-derivative at **b**.

$$f_y(a, y) =$$

$$\text{so: } f_y(a, b) =$$